

Position Based Dynamics

Matthias Müller et al.
VRIPHYS 2006

Presented by Eunki Hong

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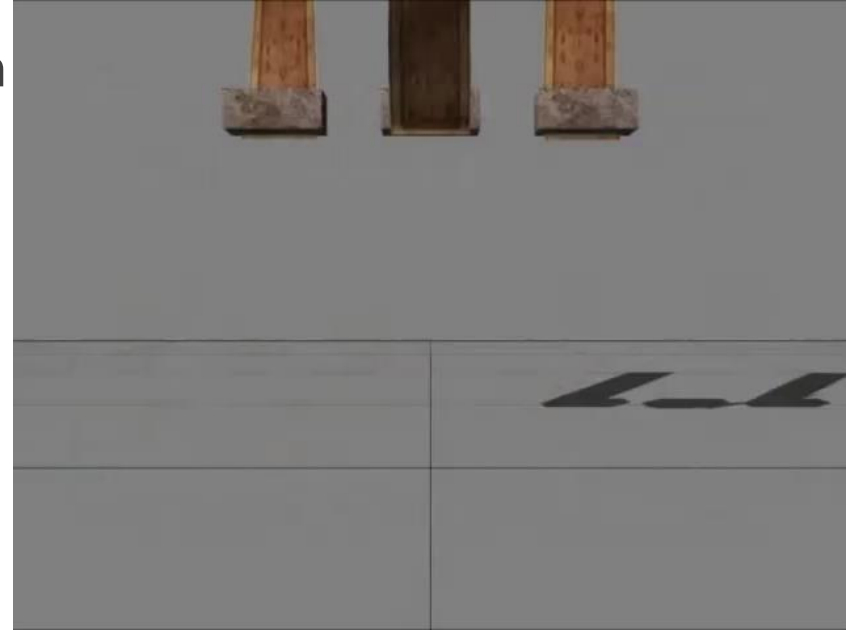
Computer Graphics @ Korea University

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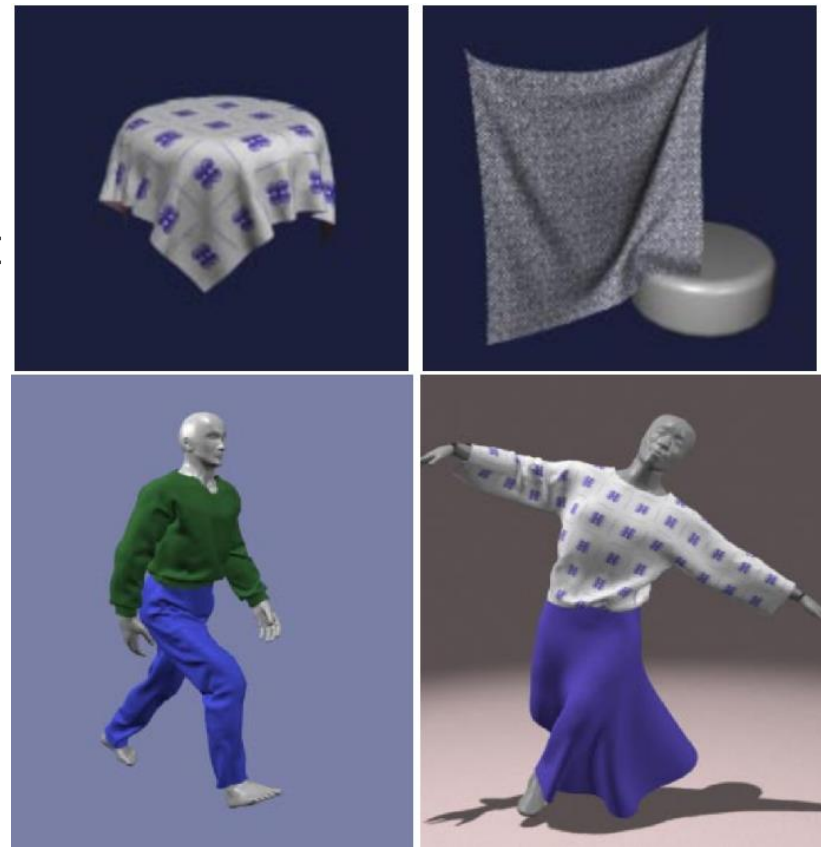
Introduction

- Traditional approach to simulation dynamic objects had been work with forces
- $F = ma \rightarrow$ Calculate accelerations
 $\rightarrow$ Error
- This paper use only constraint and position
- Main Feature of PBD
 - Control over explicit integration.
 - Removes the typical instability problems.
 - Positions of vertices can directly be manipulated during the simulation.
 - Easy to understand and implement.



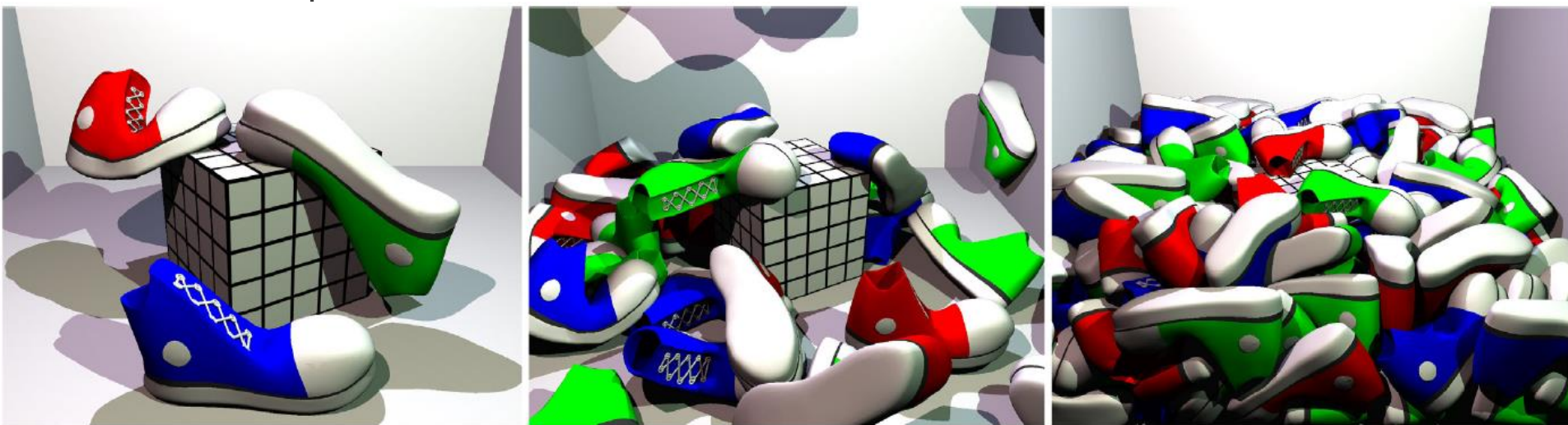
Related Work

- Large Steps in Cloth Simulation
 - [Baraff D. and Witkin A. / Proceedings of ACM SIGGRAPH 1998]
- Using constraint function
- Instead of computing constraint function energy, this paper solve for the equilibrium configuration and project positions



Related Work

- Meshless Deformations Based on Shape Matching
 - [Matthias Müller *et al* / Proceedings of ACM SIGGRAPH 2005]
- Closest method with this paper
- Only treat one specialized global constraint. Therefore, do not need a position solver.



Position Based Simulation



Algorithm Overview

- (1) **forall** vertices i
 - (2) initialize $\mathbf{x}_i = \mathbf{x}_i^0, \mathbf{v}_i = \mathbf{v}_i^0, w_i = 1/m_i$
 - (3) **endfor**
 - (4) **loop**
 - (5) **forall** vertices i **do** $\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t w_i \mathbf{f}_{ext}(\mathbf{x}_i)$
 - (6) dampVelocities($\mathbf{v}_1, \dots, \mathbf{v}_N$)
 - (7) **forall** vertices i **do** $\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$
 - (8) **forall** vertices i **do** generateCollisionConstraints($\mathbf{x}_i \rightarrow \mathbf{p}_i$)
 - (9) **loop** solverIterations **times**
 - (10) projectConstraints($C_1, \dots, C_{M+M_{coll}}, \mathbf{p}_1, \dots, \mathbf{p}_N$)
 - (11) **endloop**
 - (12) **forall** vertices i
 - (13) $\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$
 - (14) $\mathbf{x}_i \leftarrow \mathbf{p}_i$
 - (15) **endfor**
 - (16) velocityUpdate($\mathbf{v}_1, \dots, \mathbf{v}_N$)
 - (17) **endloop**
- Initialize
- Move Position
(external forces)
- Solver
(internal forces)

Algorithm Overview - Solver

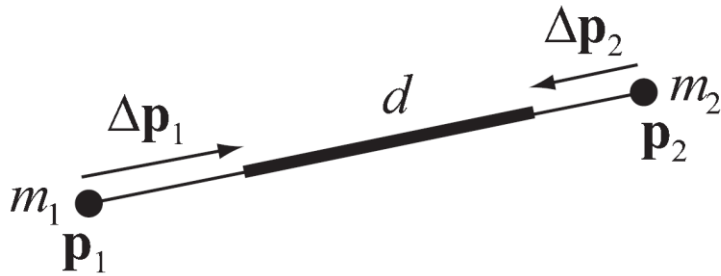
```
(9)    loop solverIterations times  
└─ (10)    projectConstraints( $C_1, \dots, C_{M+M_{coll}}$ ,  $\mathbf{p}_1, \dots, \mathbf{p}_N$ )  
    (11)    endloop  
↓  
projectConstraints( $C_1, \dots, C_{M+M_{coll}}$ ,  $\mathbf{p}_1, \dots, \mathbf{p}_N$ )  
for( $C = C_1$  to  $C_M$ )  
     $q \leftarrow C.get\_using\_vertices\_set()$ ;  
    ConstraintProjection( $C, q$ );  
endfor  
for( $C = C_{M+1}$  to  $C_{M+M_{coll}}$ )  
     $q \leftarrow C.get\_using\_vertices\_set()$ ;  
    CollisionResponse( $C, q$ );  
endfor  
end
```


Define Constraint

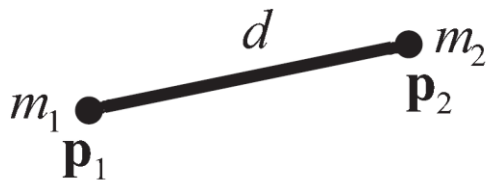
- Constraint is a function
 - $C : \mathbb{R}^{3n_j} \rightarrow \mathbb{R}$
 - n_j : the number of vertices that constraint use
- Constraints expression internal energy
- $C(p) = 0$ mean **System is stable**

Constraint Example

- $C(p_1, p_2) = \|p_1 - p_2\| - d$



$$C(p_1, p_2) = \|\Delta p_1\| + \|\Delta p_2\|$$



$$C(p_1, p_2) = 0$$

Constraint Projection

- Let define terms like
 - C : Current constraint
 - n : The number of vertices that constraint use
 - p_i : i 's vertex position vector (size is 3)
 - w_i : Inverse of i 's vertex mass (= $1/m_i$)
 - p : $(p_1, p_2, \dots, p_n)$ Connect of vertices position vector (size is $3n$)
 - Δp : next timestep vertices position (size is $3n$)
- We want to find Δp s.t. $C(p + \Delta p) = 0$
- and $\sum_i m_i \Delta p_i = 0$ (momentum conserved)

Constraint Project Equation (1/2)

- Taylor expansion

$$C(p + \Delta p) \doteq C(p) + \nabla_p C(p) \cdot \Delta p = 0$$

- Direction of Δp

$$\Delta p = \lambda \nabla_p C(p) \quad (\text{only if all vertices mass are equal})$$

- Putting together

$$\Delta p = - \frac{C(p)}{\|\nabla_p C(p)\|^2} \nabla_p C(p)$$

- For individual position

$$\Delta p_i = -s \nabla_{p_i} C(p) ,$$

$$s = \frac{C(p)}{\sum_j \|\nabla_{p_j} C(p)\|^2} \quad (\text{is scaling factor})$$

Constraint Project Equation (2/2)

- Consider mass

$$\Delta p_i = -s w_i \nabla_{p_i} C(p) ,$$

$$s = \frac{C(p)}{\sum_j w_j \|\nabla_{p_j} C(p)\|^2}$$

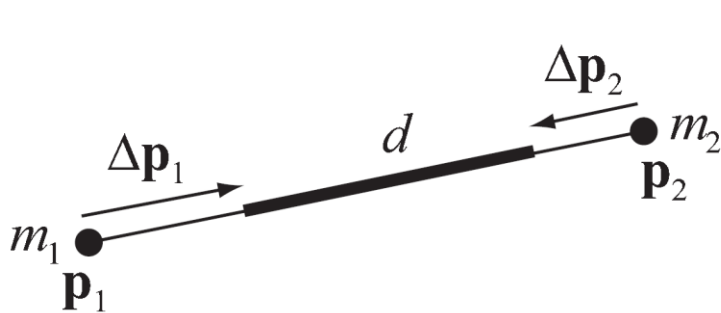
- Why?

$$\sum_i m_i \Delta p_i = 0 \text{ (linear momentum conserved)}$$

$$= -s \sum_i \nabla_{p_i} C(p) = 0 \text{ (translation invariance)}$$

Constraint Project Example

- $C(p_1, p_2) = \|p_1 - p_2\| - d$



$$\nabla_{p_1} C(p) = \frac{p_1 - p_2}{\|p_1 - p_2\|}$$

$$\nabla_{p_2} C(p) = -\frac{p_1 - p_2}{\|p_1 - p_2\|}$$

- $$\therefore s = \frac{C(p)}{\sum_j w_j \|\nabla_{p_j} C(p)\|^2} = \frac{\|p_1 - p_2\| - d}{w_1 + w_2}$$
$$\therefore \Delta p_1 = -\frac{\|p_1 - p_2\| - d}{w_1 + w_2} w_1 \frac{p_1 - p_2}{\|p_1 - p_2\|}$$
$$\Delta p_2 = +\frac{\|p_1 - p_2\| - d}{w_1 + w_2} w_2 \frac{p_1 - p_2}{\|p_1 - p_2\|}$$

Stiffness parameter of Constraint

- $p_i \leftarrow p_i + k \Delta p_i$ ($k \in [0...1]$)

(9) **loop** solverIterations **times**

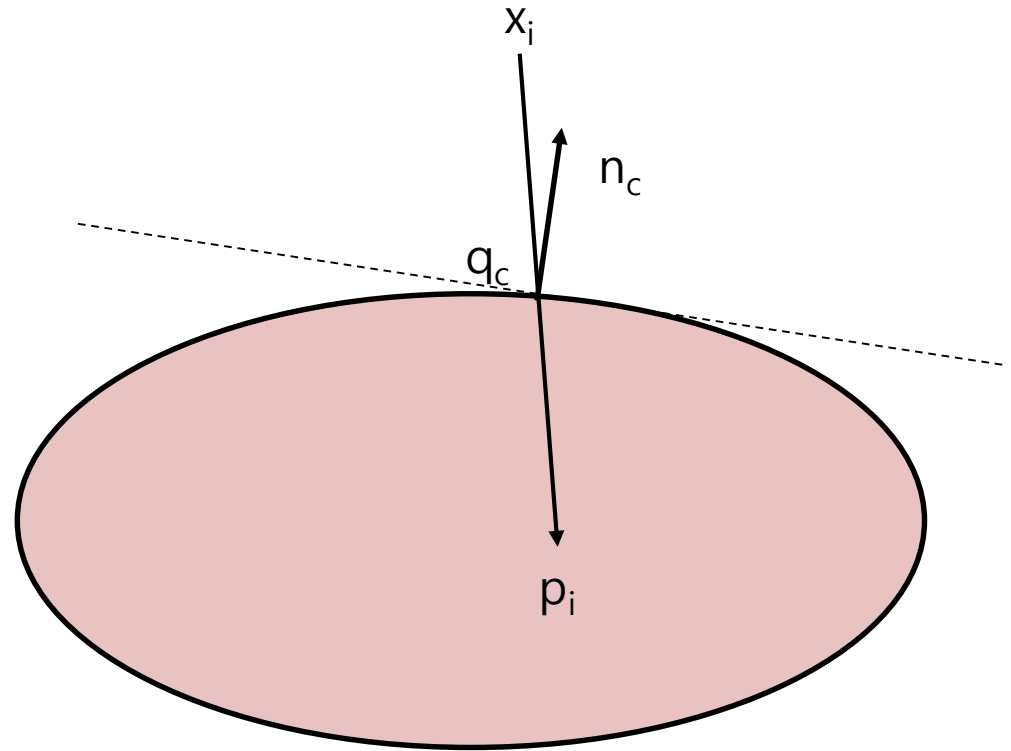
(10) projectConstraints($C_1, \dots, C_{M+M_{coll}}, \mathbf{p}_1, \dots, \mathbf{p}_N$)

(11) **endloop**

- Distance between prev point and result point is $\Delta p_i(k)$
- After n_s iteration, distance become $\Delta p_i(1 - (1 - k)^{n_s})$
- So we need to change k' dependent on iteration time
- $k' = 1 - (1 - k)^{1/\text{solverIterations}}$
 $p_i \leftarrow p_i + k' \Delta p_i$

Collision Constraint

- If $x_i \rightarrow p_i$ ray enters an object,
 - q_c : entry point
 - n_c : surface normal
- Make Constraint
 $C(p_i) = (p_i - q_c) \cdot n_c$
Stiffness $k=1$
- This constraint goal is $C(p_i + \Delta p_i) \geq 0$,
not $C(p_i + \Delta p_i) = 0$



Collision Response

- Collision constraint will work only $C(p_i) < 0$

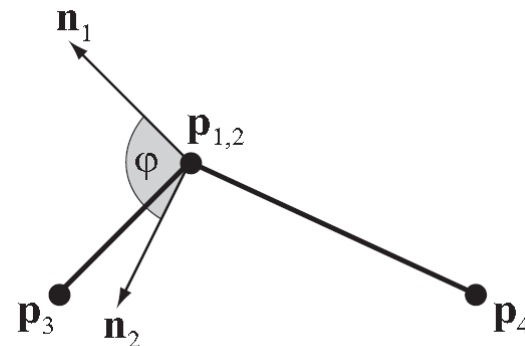
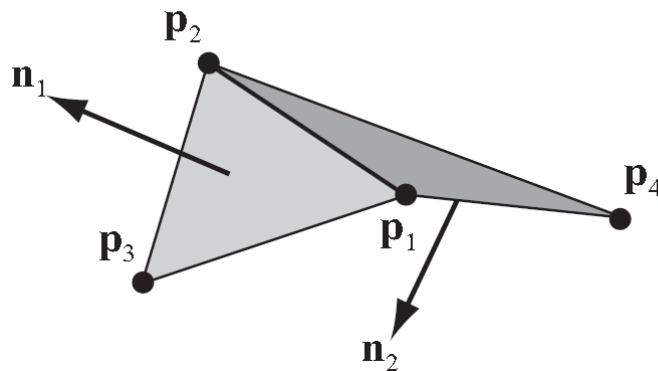
$$\Delta p_i = \begin{cases} 0 & C(p_i) \geq 0 \\ -s \ w_i \ \nabla_{p_i} C(p_i) & C(p_i) < 0 \end{cases}$$

Cloth Simulation



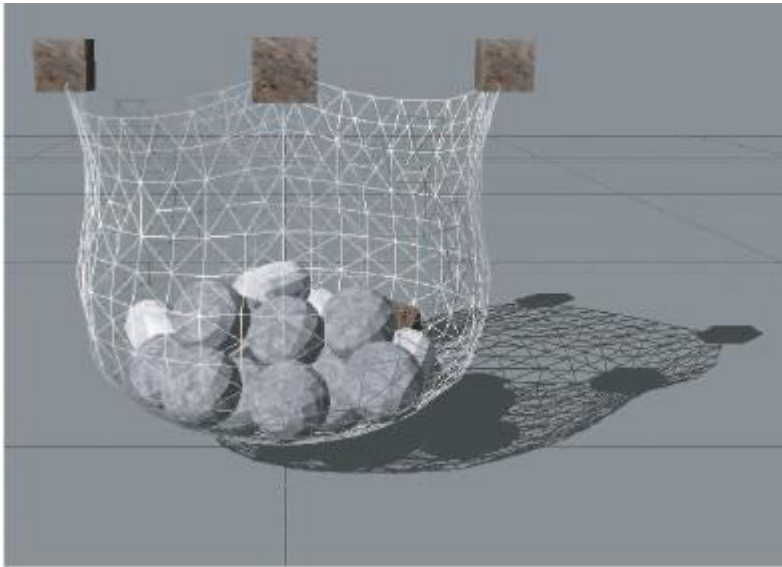
Representation of Cloth

- There are two type of internal force
- Stretch
 - $C_{\text{stretch}}(p_1, p_2) = \|p_1 - p_2\| - l_0$
 - Stiffness $k = k_{\text{stretch}}$
- Bending
 - $C_{\text{bend}}(p_1, p_2, p_3, p_4) = \text{acos}\left(\frac{(p_2 - p_1) \times (p_3 - p_1)}{\|(p_2 - p_1) \times (p_3 - p_1)\|} \cdot \frac{(p_2 - p_1) \times (p_4 - p_1)}{\|(p_2 - p_1) \times (p_4 - p_1)\|}\right) - \varphi_0$
 - Stiffness $k = k_{\text{bend}}$

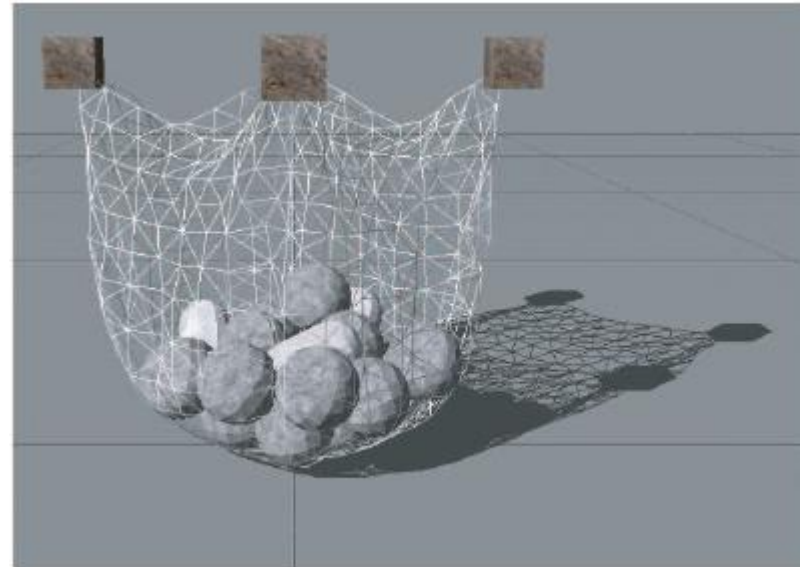


Bending Constraint

With Bending

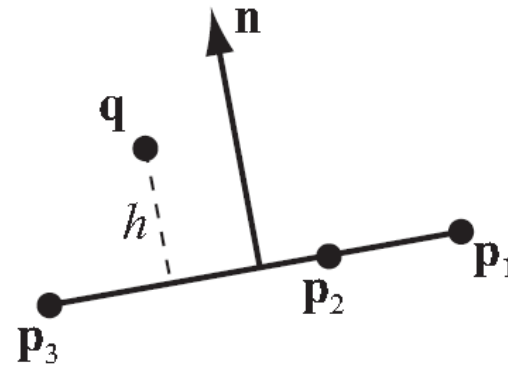
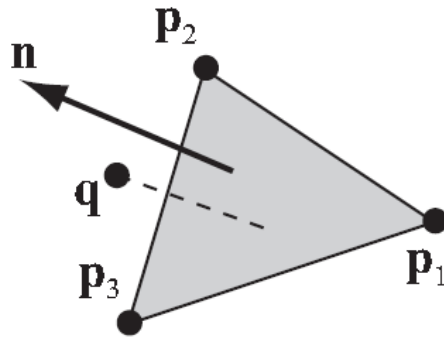


Without Bending



Self Collision

- If vertex q move through a triangle p_1, p_2, p_3



$$C(q, p_1, p_2, p_3) = (q - p_1) \cdot n - h \quad \text{and}$$

$$C(q, p_1, p_2, p_3) = -(q - p_1) \cdot n - h$$

- When
 - n : normal vector of triangle p_1, p_2, p_3
 - h : cloth thickness

Cloth Balloons

- $C(p_1, \dots, p_N) = \left(\sum_i (pt_1^i \times p_{t_2}^i) \cdot p_{t_3}^i \right) - k_{\text{pressure}} V_0$
 $\perp$ (Volume of mesh)



- If $k_{\text{pressure}} > 1$ mesh will get overpressure
- $\nabla_{p_i} C(p) = \sum_{j:t_1^j=i} (pt_2^j \times p_{t_3}^j) + \sum_{j:t_2^j=i} (pt_3^j \times p_{t_1}^j) + \sum_{j:t_3^j=i} (pt_1^j \times p_{t_2}^j)$
 $= \sum(\text{normal vector who contain } p_i) * (\text{area of flat})$

Position Based Dynamics

paper #11

**All sequences
captured in real time**